Quadratic equations

1 $x = 2$

2

a $a = -10$ **b** $a = 1$

3

a $m = 0, 14$ **b** $m = 5, -4$ **c** $m = 6, 1$ **d** $m = 6, 5$

e $m = 14, 13$ **f** $m = 0, 10$ **g** $m = 0, -2$ **h** $n = -4, -2$

i $y = 0, \frac{1}{3}$ **j** $y = 0, \frac{1}{5}$ **k** $y = 7, 1$ **l** $y = -1, 3$

m $b = -\frac{5}{3}, \frac{3}{4}$ **n** $k = 1, \frac{7}{6}$

4

a $x = \pm 8$ **b** $x = 0, -12$ **c** $x = -7, -6$ **d** $x = -7, 3$

e $x = 4, 10$ **f** $x = 10, -5$ **g** $x = 11$ **h** $x = 3, -2$

i $x = -8, -10$ **j** $x = 6, -8$ **k** $x = 2, -10$ **l** $x = 19, -6$

m $x = -\frac{5}{3}, 4$ **n** $x = -\frac{2}{5}, -4$ **o** $x = \frac{9}{4}, 4$ **p** $x = \frac{8}{5}, 9$

q $x = -\frac{1}{2}, 2$ **r** $x = \frac{3}{2}, 2$ **s** $x = \frac{5}{3}$ **t** $x = \frac{3}{5}, 2, \frac{5}{2}, 4$

5

a $x = 4, -7$ **b** $x = 8, -3$ **c** $x = 7, -4$ **d** $x = 9, -5$

e $x = -\frac{3}{2}, 3$ **f** $x = -\frac{5}{2}, 12$ **g** $x = -\frac{4}{3}, 5$ **h** $x = -7, 3$

i $x = 1, -\frac{7}{3}$ **j** $x = \frac{1}{2}, -\frac{5}{2}$

Applications

6 $x = \pm 4$

7

a $x = 2, -3$ **b** $x = 3, -7$

8

a $t = 3, 9$ **b** 3 hours **c** 6 hours

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